

Ziwei Gu

CONTACT INFORMATION

Email: ziweigu@g.harvard.edu
Website: <https://www.ziweigu.com/>

RESEARCH INTERESTS

Human-Computer Interaction (HCI)
Natural Language Processing, Machine Learning, Human-AI Interaction

EDUCATION

Harvard University, Cambridge, Massachusetts
Ph.D. Computer Science, August 2022 – Present
Advised by Elena Glassman, Assistant Professor of Computer Science
Cornell University, Ithaca, New York
B.A. Computer Science, Magna cum laude, August 2017 – December 2020
B.A. Mathematics, August 2017 – December 2020

PEER-REVIEWED PAPERS

Ziwei Gu, Joyce Zhou, Ning-Er (Nina) Lei, Jonathan K. Kummerfeld, Mahmood Jasim, Narges Mahyar, and Elena L. Glassman. [AbstractExplorer: Leveraging Structure-Mapping Theory to Enhance Comparative Close Reading at Scale](#). In *ACM Symposium on User Interface Software and Technology (UIST)*, 2025.

Ziwei Gu, Ian Arawjo, Kenneth Li, Jonathan K. Kummerfeld, and Elena L. Glassman. [An AI-Resilient Text Rendering Technique for Reading and Skimming Documents](#). In *ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2024.

Katy Ilonka Gero, Chelse Swoopes, **Ziwei Gu**, Jonathan K. Kummerfeld, and Elena L. Glassman. [Supporting Sensemaking of Large Language Model Outputs at Scale](#). In *ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2024.

Ziwei Gu, Owen Raymond, Naser Al Madi, and Elena L. Glassman. [Why Do Skimmers Perform Better with Grammar-Preserving Text Saliency Modulation \(GP-TSM\)? Evidence from an Eye Tracking Study](#). In *ACM CHI Conference on Human Factors in Computing Systems Late Breaking Work (CHI LBW)*, 2024.

Ziwei Gu^{*}, Gauri Jain^{*}, Hongwen Song^{*}, Isak Diaz^{*}, Margaux Masson-Forsythe^{*}, and Jorge Valdes. [BiomeAzuero2022: A Fine-Grained Dataset and Baselines for Tree Species Classification with Ground Images](#). In the *37th AAAI Conference on Artificial Intelligence (AAAI-23) AI for Social Good Workshop*, 2023.

Jing Nathan Yan, **Ziwei Gu**, and Jeffrey M Rzeszotarski. [Tessera: Discretizing Data Analysis Workflows on a Task Level](#). In *ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2021.

Ziwei Gu^{*}, Jing Nathan Yan^{*}, and Jeffrey M Rzeszotarski. [Understanding User Sensemaking in Machine Learning Fairness Assessment Systems](#). In *WWW'21: The Web Conference (WWW)*, 2021.

Jing Nathan Yan, **Ziwei Gu**, Hubert Lin, and Jeffrey M Rzeszotarski. [Silva: Interactively Assessing Machine Learning Fairness Using Causality](#). In *ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2020.

^{*}: Equal contribution

RESEARCH
EXPERIENCE

- | | |
|---|--------------|
| Harvard University , Cambridge, Massachusetts | 2022-Present |
| <ul style="list-style-type: none"> Developed and evaluated an LLM-powered recursive sentence compression algorithm for grammar-preserving extractive summarization and text saliency modulation visualization, which led to significantly more efficient reading comprehension compared to existing methods. Created and evaluated an LLM-powered interactive system for analyzing research abstracts at scale (>1,000); validated through ablation and summative studies and demonstrated significant improvements in reading and skimming efficiency on a large research corpus. | |
| Cornell University , Ithaca, New York | 2019-2020 |
| <ul style="list-style-type: none"> Adapted the Transformer model for open information extraction, modeled as a sequence-to-sequence transduction task. Our model was competitive with state-of-the-art systems but did not rely on other NLP tools. | |

INDUSTRY
EXPERIENCE

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| Lyft , San Francisco, California | |
| <i>Data Scientist Intern</i> | 2020 |
| <ul style="list-style-type: none"> Analyzed millions of rider payment transactions; applied clustering to identify user profiles and proposed a Lyft Family feature, which was implemented and remains active in the app. Designed a Markov chain model to study driver churn and estimated transition probabilities between active and churn states; quantified the statistical significance of incentive effects and delivered insights that informed driver engagement strategies. | |

TEACHING
EXPERIENCE

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|---|-----------|
| <i>Head Teaching Fellow</i> , COMPSCI 178 Engineering Usable Interactive Systems, Harvard CS | 2023 |
| <i>Graduate Teaching Assistant</i> , CS 4410 Operating Systems, Cornell CS | 2021 |
| <i>Teaching Assistant</i> , CS 4780 Machine Learning, Cornell CS | 2019-2020 |
| <i>Teaching Assistant</i> , CS 3410 Computer System Organization and Programming, Cornell CS | 2020 |
| <i>Teaching Assistant</i> , CS 2110 Object-Oriented Programming and Data Structures, Cornell CS | 2018-2019 |

OTHER
EXPERIENCE

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| <i>Treasurer</i> , Harvard Chinese Students and Scholars Association (HCSSA) | 2022-Present |
| <i>Fellow (Intellectual and Cultural)</i> , The Student Center at Harvard Griffin GSAS | 2023-2025 |
| <i>Project Lead</i> , Statistics Faculty Award winner, Cornell Data Science Team | 2018-2020 |
| <i>Resident Advisor</i> , Clara Dickson Hall, Cornell University | 2019-2021 |

Latest update: 12/2025